

Answer Key — Laws of Reflection

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

Q1 (b) **normal** — angles are measured from the normal.

Q2 (c) **$i = r$** — the angle of incidence equals the angle of reflection.

Q3 (b) **converge** — a concave mirror converges parallel rays.

Q4 (c) **0°** — along the normal, $i = r = 0^\circ$.

Q5 (a) **Both true, R explains A** — the curve spreads the beam while each ray obeys $i = r$ — as R explains.

Q6 **plane** — all three lie in one plane (second law).

Q7 **converges** — a concave mirror converges parallel rays.

Short answer

Q8 (1) The angle of incidence equals the angle of reflection ($i = r$). (2) The incident ray, the normal and the reflected ray all lie in the same plane.

Q9 Plane: stay parallel; concave: converge; convex: diverge.

Q10 Incident ray: ray hitting the mirror; reflected ray: ray bouncing off; normal: line at 90° to the mirror at the point of incidence.

Long / application

Q11 Shine a thin beam (through a single comb slit) onto a plane mirror on white paper. Draw the incident ray, reflected ray and the normal at the point of incidence. Measure i and r with a protractor; repeat at several angles. The angles are found equal each time, verifying $i = r$.

Q12 A concave mirror converges parallel sun rays to a point, concentrating light and heat there (enough to heat liquids or even burn paper) — useful for a solar concentrator. A convex mirror diverges the rays, spreading them out, so it cannot concentrate sunlight.
